

R00065426 – Natural Language Processing – Assignment 1

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Abstract

This is my submission for assignment 1 in the Natural Language Processing module of the MSc. in AI.

1. Introduction & Task Description

This assignment for the Natural Language Processing module seeks to uncover biases against regions and languages that are under-represented in the LLM data sets. Most of the data that these LLMs are built on is from the Internet, and greater than 50% of all internet content is in the English language [1]. Therefore, we should expect a bias towards information that was originally stored in English, and away from other languages. Underrepresented languages should be a rich source of information about the culture that those languages emerged from, providing information about local/regional culture and traditions. What is uncertain is whether this culture specific cultural knowledge is available through interrogation of the LLMs that are bias against these underrepresented languages/cultures/traditions. This assignment seeks to quantify the extent to which these biases exist.

All of the code and datasets used for this project can also be found on github.

https://github.com/paulnagle/nlp_assignment_1

The file in colab can be found at this link

https://colab.research.google.com/drive/1t-kJR_2uix8BBmlfqSTX_dqXt4E1ZsQ9?usp=sharing

2. Dataset and locale selection

I re-used some the datasets from [BLEnD \(Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages\)](#) which is what this current assignment was inspired by.

BLEnD [2] provided a curated list of both multiple choice and single answer questions for multiple locales, all with correct answers provided. I would not have been able to produce such a dataset independently.

The assignment required us to choose 4 locales including at least one non-Latin script locale, and

at least one low resource locale. The locales that I chose were

- **en-US** (English-US): High-resource baseline
- **en-GB** (English-UK): High-resource, European locale
- **zh-CN** (Chinese-China): Non-Latin script, major language
- **am-ET** (Amharic-Ethiopia): Low-resource, under-represented locale

The model that I chose to use was mistralai / Mistral-7B-Instruct-v0.3 [3] from huggingface which is an instruct fine-tuned version of the Mistral-7B-v0.3 [4] suitable for question-and-answer scenarios.

3. Methodology

3.1 Baseline

I implemented a baseline system for both Track A Short Answer Questions (SAQ) and Track B (MCQ). Both tracks implement a direct prompt-to-model inference pipeline, with results compared to the dataset given answers, after appropriate normalisation and lemmatization.

3.1.1 Short Answer Question methodology

Questions and answers were loaded from datafiles that I borrowed from the BLEnD project that the assignment was based on. I used a sample size of 500 questions per locale.

I used persona-based prompts to ask the questions i.e. "You are an American cultural expert with deep knowledge of US everyday life, food, traditions, and customs".

The questions were also sent to the LLM with instructions to use a maximum of 100 tokens in the answer. This influenced the LLM to be brief in it's responses.

The answers were normalised with locale specific operations, including a specialised python library (AmharicNLP [5]) for the Ethiopian Amharic script.

Answers were generated with Temperature set to 0 for reproducibility, and all answers returned a token probability from the LLM which was used to calculate an overall confidence score for the questions. An abstention method was implemented which would discard any answer

with a confidence score lower than 0.3, but none were encountered. Confidence scores were values i.e. between 0.0 and 1.0

The answers were evaluated to be correct based on some basic string searches after normalisation. Exact matches or substring matches were accepted.

I then calculated some metrics, accuracy, confidence analysis i.e. confidence gap between correct and incorrect answers.

3.1.2 Multiple Choice Qs methodology

For the multiple-choice question evaluations, the data was again borrowed from the BLEND project. I took a sample set of 43,000 questions in total, distributed across the locales. Lacking the regional information and language skills, I would not have been able to generate these datasets myself.

Each question from the dataset included a prompt (with json response requirement), 4 answer choices (A, B, C, D), the correct answer and country associations for each choice.

The prompts being sent to the LLM included the prompt from the dataset, and extra locale-persona based prompt details.

The prompts were also sent with max tokens set to 50. This influenced the LLM to respond in the brief format required.

To extract the answer details from the LLM response, I used json to parse the response (responses were required to be of the format: {"answer_choice": "A"}). The LLM did not always obey the strict requirement for the answer format, so I added fallback of regex searches for A/B/C/D.

Evaluating the answers was easier than for the SAQ dataset. The LLM for the most part stuck to the response format requested, and parsing the answers was performant.

3.1.3 Normalisation approach

The implemented normalization approach was to normalize the text before matching generated answers against the reference answers. All inputs are first given a standard cleanup i.e. Unicode NFC normalization, removal of any control characters, standardizing line breaks, cleaning up whitespace and converting everything to lowercase.

For **en-US** and **en-UK** the normalization was identical, as they are both English in latin script. I removed all punctuation except for apostrophes and hyphens, both of which can contribute to the meaning of the word being normalised. I then lemmatized each token with NLTK WordNet. First as nouns and then as verbs. Lemmatization in this case preserves meaning better than stemming, so I chose not to implement any stemming.

For **zh-CN**, the normalization process was different. I removed all latin text (except numbers) and all Chinese punctuation. Whitespace is not a reliable boundary in Chinese, so I removed all whitespace entirely.

For **am-ET**, I used an NLP toolkit specially designed for the locale's script, called "amharicNLP" [5]. Ethiopian script is called Amharic and is a low resource language, even though there are close to 60 million native speakers. I ran the methods provided by the library to standardize the characters, clean punctuation spacing (Amharic uses unique punctuation marks like '፡', '፣', '፥', '፦', '፧', '፨'), to remove high frequency stop/filler words and to convert each word to its canonical dictionary form.

3.1.4 Experimental setup

The python notebook was run in Google colab using a H100 GPU. I implemented a fallback mode on the local development, so that the code could still be tested and debugged on a computer that couldn't run the larger 7B models.

I configured 4-bit quantization for the Mistral-7B LLM, to reduce the amount of memory required and hopefully to improve the efficiency of the inference (with lower memory bandwidth required). I would have liked to experiment more with different quantisation settings, but due to time constraints I was unable to.

3.2 Improvements

3.2.1 Persona based prompting

For both the SAQ and MCQ tracks I implemented an extra level of persona-based prompting. This was as basic as adding some text in English to the prompts that were used. The prompts instructed the LLM to respond as a person who was an expert in the local traditions i.e. "You are an American cultural expert with deep knowledge of US everyday life, food, traditions, and customs".

The exact text of this prompt was just a free form guess at the best way to influence the LLM in natural language form. It is unclear to me how much of a difference that these persona-based prompts made to the response that I got from the LLM. If given more time, this would be an interesting area to explore. Perhaps there is a better way to define a persona from which an LLM should generate an answer.

3.2.2 Confidence estimation/abstention

The confidence estimation was calculated using a token level probability averaging approach. The generation requests to the LLM were sent with "return_dict_in_generate=True" which meant that we receive extra details in the response, such as the list of logits for each generation step. From this list I was able to use the torch library softmax functionality to obtain the selected token's

probability and I was able to average these to calculate the overall probability for each token. The final confidence score was then the arithmetic mean of all the probabilities. [6]

I was then able to produce a variety of metrics that would potentially be useful for analysis. i.e. Average confidence (all), Average confidence (correct), Average confidence (incorrect), Confidence gap, Abstention rate, Coverage, Precision on answered.

One thing to note here is that as the following settings were always applied:

"temperature=0.0" and "do_sample=False", then the highest probability token was always returned by the LLM (greedy decoding), which potentially inflated the confidence scores.

I also implemented an abstention mechanism where answers with a confidence level of less than 0.3 would have been rejected.

In practice, the confidence levels returned were always quite high (possibly due to the greedy decoding) even with incorrect answers, so the utility of my confidence estimation is questionable, and the abstention method was never triggered.

3.2.3 Colab/local dev setup

During the repetitive work of running and debugging the notebook, and perhaps due to my lack of experience with Jupiter notebooks and Google colab, it became apparent that I would not be able to work solely in colab and that I would have to split the work between my laptop and colab. To this end I implemented code that allowed the notebook to be run in both environments with no code changes.

If the notebook was being run on colab, then I would mount my google drive with the data sets, and I would use the colab GPUs to run a large 7B model.

If I was running the notebook on my laptop, then I would use the datasets on the local filesystem, run a very small subset of tasks, and use a very small LLM that ran on the CPU on my laptop (M2 MacBook Air 16gb ram).

4. Evaluation

4.1 Overall Performance

Overall, it can be seen from the results that the least represented language performed by far the worst of all of the languages tested. In comparison with the MCQ results (detailed later) on approximately the same topics, but asked in the native language, there is a huge difference. Questions asked about Ethiopian culture and

traditions. are answered much better when they are asked in English.

The evaluations of the SAQ and MCQ tracks are listed below

4.2 SAQ Evaluation Summary

Locale	Correct	Total	Accuracy
en-US	392	500	78.4%
en-GB	375	500	75.0%
zh-CN	241	500	48.2%
am-ET	12	500	2.6%
Overall	1021	2000	51.0%

These SAQ questions were all asked in the script of the country. The most striking result is the 2.6% accuracy rate for the questions that were asked in the Amharic script.

4.2.1 Confidence Statistics

Answer Confidence Averages	en-US	en-GB	zh-CN	am-ET
All	0.826	0.820	0.776	0.852
Correct	0.833	0.825	0.777	0.853
Incorrect	0.803	0.806	0.775	0.852
Gap	0.020	0.019	0.002	0.001
Abstained	0	0	0	0

Average confidence (all): 0.818

Average confidence (correct): 0.817

Average confidence (incorrect): 0.820

Confidence gap (correct - incorrect): -0.003

As noted previously, these confidence statistics are skewed significantly by the temperature = 0 setting. Nevertheless, relative to each other the LLM model was still reporting very high levels of confidence with consistently incorrect answers, in the case of the am-ET locale. There was very little difference in the confidence levels between the correct answers and the incorrect answers. This

leads us to believe that the Large Language Model had no real idea if the information that it was presenting was correct or incorrect. It was blindly confident about all of its outputs. So much so that the abstention level of 0.3 was never reached, even once. We could have set the abstention level much higher with the same result.

4.3 MCQ Evaluation Summary

Locale	Correct	Total	Accuracy
en-US	8013	9273	86.4%
en-GB	9330	10750	86.8%
zh-CN	8947	10625	84.2%
am-ET	7497	10750	69.7%
Overall	33787	41398	81.6%

Best performing locale: **86.8%**

Performance gaps from best:

en-US: -0.4%
en-GB: 0.0%
zh-CN: -2.6%
am-ET: -17.1%

Answer Confidence Averages	en-US	en-GB	zh-CN	am-ET
All	0.943	0.949	0.951	0.955
Correct	0.946	0.950	0.953	0.958
Incorrect	0.923	0.938	0.941	0.946
Gap	0.023	0.013	0.012	0.012
Abstained	0	0	0	0

Average confidence (all): 0.950

Average confidence (correct): 0.952

Average confidence (incorrect): 0.937

Confidence gap (correct - incorrect): 0.015

Interestingly, we can see that the lower resourced locale achieved much better results when the questions were asked in English. This raises some concerns that I will examine in the Responsible AI Reflection section below.

5. Error Analysis

5.1 Failure examples (10 - 12)

5.1.1 SAQ Sample Failures

5.1.1.1 en-US

Here is a typical failing question from en-US

"14. ID: Ji-ko-40

Question: What do high school students typically do during break time in schools in the US?

Generated: they socialize study or participate in school activity

Expected: eat lunch, check facebook, play on phones

So this is a nuanced failure. It would be easy to rank the generated answer as correct, in certain circumstances

5.1.1.2 en-GB

"15. ID: Ji-ko-40

Question: What do secondary school students typically do during break time in schools in the UK?

Generated: eat socialize or play sport

Expected: eat lunch, work part-time, go outside"

Again, this is a nuanced failure which could easily be considered correct, given the appropriate circumstances.

5.1.1.3 zh-CN

"12. ID: Al-en-43

Question: 在中国，家庭中流行的室内活动是什么？

Generated: 在中国家庭中流行的室内活动是吃饭看电影打牌读书唱歌练习舞蹈等 在中国家庭中流行的室内活动是吃饭电影打牌读书唱歌舞蹈等

Expected: 麻将, 看电视, 斗地主"

Google translate offers this translation of the question and answer

"Question: What are some popular indoor activities in Chinese families?

Generated: Popular indoor activities in Chinese families include eating, watching movies, playing cards, reading, singing, and dancing.

Expected: Mahjong, watching TV, Dou Dizhu (a popular Chinese card game)."

Again, the difference between the generated answer here and the expected answer is nuanced. We have "watching movies" not matching with "watching tv".

Ironically, probably the best technology for comparing equivalence of these types of things would be another LLM. An alternative implementation of the answer checker would have been to use a separate model to test for approximate equivalence between the generated and expected answers.

5.1.1.4 am-ET

“81. ID: Ji-ko-40

Question: በኢትዮጵያ ሁለተኛ ደረጃ በትምህርት ቤቶች ውስጥ በአረፍት ጊዜ ተማሪዎች ምን ያደርጋሉ?

Generated: በኢትዮጵያ ሁለተኛ ደረጃ በትምህርት ቤቶች ውስጥ በአረፍት ጊዜ

Expected: ኢስ ይጫወታሉ, ሻይ ይጠጣሉ, ከጓደኛ ጋር ያወራሉ”

Google translate gives me “What do students do during recess in secondary schools in Ethiopia?

Generated: During recess in secondary schools in Ethiopia

Expected: They play football, drink tea, talk to friends”

This might be a failure in the NLP processing library for the Amharic script; it could be a failure in the LLM returning a truncated sentence or it could be a failure in my code to display the generated results. Perhaps 100 tokens were not enough to capture the Unicode for the generated Amharic script.

5.1.2 MCQ Sample Failures

The dataset of questions that I was using for these questions has some unusual characteristics that I was not fully able to understand. The same questions were asked multiple times, with different IDs. For example, the question “ID: AI-en-01_158

Question: What is a common snack for preschool kids in the UK? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

is asked multiple times, with slight variances in the answers provided i.e.

“A. apple

B. chocolate paste

C. jam sandwiches

D. semolina”

and

“A. apple

B. chocolate paste

C. jam sandwiches

D. potato fries”

and

“A. apple

B. chocolate paste

C. jam sandwiches

D. noodles”

It was unclear to me of the exact utility of this approach, but it was apparent that if the LLM got one of these questions wrong, then it was likely to get a larger batch of the same question wrong as well.

A number of the questions also had the unusual phrase “(Provide Arabic numerals (e.g., 20) only.)” in the text.

5.1.2.1 en-US

“ 2. ID: AI-en-01_170

Question: What is a common snack for preschool kids in the US? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

A. chocolate paste

B. cookie

C. egg

D. fruit

Answer:

Predicted: B | Correct: D

Confidence: 0.913”

Here we see an example of an incorreced answered MCQ question. The annotated answer to the question was “fruit” but the model predicted that “cookie” would be the correct answer. This is a debatable failure, many schoolchildren in the US would presumably get a cookie for a snack during the day, so it should not be considered to be 100% incorrect.

There are many socio-economic influences that would potentially take effect here. Parents education levels, employment status, location, school type, income bracket etc etc.

Yet, the model returns a confidence level of 0.913 out of 1.0, so very high levels of confidence. The confidence calibration of the LLM seems to be set very high.

5.1.2.2 en-GB

"1. ID: AI-en-01_3

Question: What is a common snack for preschool kids in the UK? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

- A. chocolate paste
- B. cookie
- C. egg
- D. fruit

Answer:

Predicted: B | Correct: D

Confidence: 0.934"

The same questions again, but this time for the en-GB locale and asked for UK children. The same results were returned. The term cookie to my mind is an American phrase. UK residents would be more likely to use the term "biscuit". But the LLM still predicted "cookie" with a very high confidence level, 0.934.

This could in some ways be considered an instance where the US cultural influence has overwhelmed what should be a UK centered question and answer. That being said, this is again nuanced, and the term "cookie" while maybe not as well used, is still well understood in the UK.

5.1.2.3 zh-CN

"ID: AI-en-09_390

Question: What is a popular snack at an amusement park in China? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

- A. bombing the giant squid
- B. cake
- C. cotton candy
- D. tea

Answer:

Predicted: C | Correct: A

Confidence: 0.937"

The correct answer of "Bombing the giant squid" seems like a bizarre one to someone not familiar

with Chinese culture, but this is in fact a popular street food vendor in China

5.1.2.4 am-ET

Sticking with the same question, I notice that "potato fries" is given as a correct answer in the annotated answer. Doing some basic googling, I can see that while potato fries are not an ancient, traditional dish in Ethiopia, it is a very popular modern dish that is often consumed [7], typically tossed with spicy berbere seasoning, garlic, lemon juice, and herbs for a spicy, tangy twist!

" 1. ID: AI-en-01_957

Question: What is a common snack for preschool kids in Ethiopia? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

- A. cookie
- B. egg
- C. fruit
- D. potato fries

Answer:

Predicted: C | Correct: D

Confidence: 0.976"

Potato fries are the annotated answer to this question, presumably answered by someone familiar with the area. The LLM has chosen "fruit" over "potato fries". Fruit consumption is very low in Ethiopia [8] due to affordability with households needing to spend 11% of income to meet daily requirements and weak dietary habits [9].

This is certainly an instance of a culturally incorrect answer. The LLM has seemed to blindly super-impose Western/American ideals onto the answer, while still retaining a high internal confidence that it is giving the answer with the highest probability of being correct.

6. Responsible AI reflection

It is estimated that 1 in 6 people worldwide are using generative AI, with Ireland at number 4 in the world for usage with 44.6% of adults using it [10] The same study estimates that usage in large swaths of Africa is less than 10%. This imbalance is not really a function of AI itself but reflects the state of the worldwide economy. It seems that the inequalities of the world are transferred onto AI usage.

Similarly, the macro-economics of the world have dictated that the majority of generative AI development work is being focused in the already over privileged parts of the world. What should we

do with such advantage, those of us that are lucky enough to be a part of the new wave of development and professional interest of the behind the scenes workings of new large language model?

We have seen in some of the example answers from the project undertaken in this assignment, that western and more specifically American influences have already started to overwhelm under-resourced and under-represented cultures and traditions of areas where the global economy has not shined it's brightest lights. Bringing awareness of this situation and highlighting that the inequalities exist and that they will not serve the under-represented minorities is maybe a first step. Maybe we are still early enough in the rapid phase of huge growth in the area to be able to make small differences, and plant seeds of understanding that will raise awareness of the situation. No right-minded person will want American culture to take over the world and destroy the rich cultures and traditions of all of the countries in the world. Even though the westernisation of the world is not a new phenomenon, maybe we can agree that the logical conclusion of allowing it to propagate unchecked is not ideal, and that steps could and should be taken to mitigate it.

7. Limitations and future work

Limitations: As can be seen from the failure analysis section, many of the answers that were deemed to be incorrect, were in fact versions of the correct answers with nuanced changes. Ideally, all the answers would have been reviewed by a person knowledgeable in the area. Obviously, this would have been a large undertaking, as there were many questions, but I feel that the failure rate could have been mitigated with this approach. Conversely, all the correct answers would need to be double checked as well.

The persona-based prompting that I implemented seemed to me to be a very naïve approach. Simply, telling the LLM that it was an expert in the area does not necessarily make the LLM an expert in the area. A better approach might have been RAG based, where detailed correct information in the form of data files would have been fed to the LLM to provide it with accurate information upon which to base it's answers. The potential issue here would be that we would be spoon feeding answers to the LLM and thus biasing the results, so a degree of caution would have been needed.

I attempted to implement some quantisation/gpu usage optimisation as part of the code by including the following quantisation settings:

```
# GPU: Use 4-bit quantization
```

```
quantization_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_use_double_quant=True,
)
```

The settings that I chose in practise reduced the amount of RAM that the GPU used to 16GB. I did not have the time to experiment with other settings. It may well be possible that I could have improved throughput by adjusting these settings for the particular GPU that I used in colab. The H100 GPU had 64 GB of ram, but only 16GB were used in the run. Potentially there might have been scope to run 4 separate threads of execution through the GPU, reducing the processing time by a factor of 4.

The confidence estimation with temperature set to zero seemed to be problematic. Temperature of zero always meant that the LLM returned the most probable token and skewed the confidence estimation. I would have liked more time to investigate other methods of confidence estimation, ones that potentially did not rely on what the LLM reported back.

8. References

- [1] https://en.wikipedia.org/wiki/Languages_used_on_the_Internet
- [2] Myung, Junho, et al. "Blend: A benchmark for llms on everyday knowledge in diverse cultures and languages." *Advances in Neural Information Processing Systems* 37 (2024): 78104-78146.
- [3] <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>
- [4] <https://github.com/mistralai/mistral-inference>
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- [6] <https://aclanthology.org/2024.naacl-long.366.pdf#:~:text=Logit,readily%20adapt%20to%20scenarios%20involving>
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